**What is Casting?**

**Casting** means converting a value from one **data type** to another.
Python lets you change data types using **built-in functions** like:

- int() → convert to integer
- float() → convert to float
- str() → convert to string
- complex() → convert to complex number

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### 🧩 1. Convert to Integer → int()

Converts numbers or numeric strings to integers.

🔹 It **removes decimals** (not rounds).

```
x = int(3.8)       # float → int → 3
y = int("10")      # string → int → 10
# z = int("10.5")  # ❌ Error: cannot convert float string directly

print(x, type(x))   # 3 <class 'int'>
print(y, type(y))   # 10 <class 'int'>
```

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### 🧮 2. Convert to Float → float()

Converts integers or numeric strings to decimals.

```
a = float(5)       # int → float → 5.0
b = float("7.2")   # string → float → 7.2
c = float("10")    # string (int value) → float → 10.0

print(a, type(a))   # 5.0 <class 'float'>
print(b, type(b))   # 7.2 <class 'float'>
```

print(c, type(c))  # 10.0 <class 'float'>

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### ⚡ 3. Convert to String → str()

Converts any data type into a string.

x = str(123)  # int → str → "123"

y = str(45.67)  # float → str → "45.67"

z = str(True)  # bool → str → "True"

print(x, type(x))  # 123 <class 'str'>

print(y, type(y))  # 45.67 <class 'str'>

print(z, type(z))  # True <class 'str'>

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### 🔹 4. Convert to Complex → complex()

Converts numbers into **complex** (a + bj) form.

x = complex(5)  # int → complex → (5+0j)

y = complex(3.2)  # float → complex → (3.2+0j)

z = complex(2, 4)  # real + imaginary → (2+4j)

print(x, type(x))  # (5+0j) <class 'complex'>

print(y, type(y))  # (3.2+0j) <class 'complex'>

print(z, type(z))  # (2+4j) <class 'complex'>

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### 🚫 Invalid Conversions

Some conversions are not allowed:

# int("abc")  ❌ Error – cannot convert letters to number

# float("abc")  ❌ Error

# int("10.5") ❌ Error – must be numeric integer string

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📜 **Summary Table**

| Function | Converts To | Example | Result |
|---|---|---|---|
| int(x) | Integer | int(3.8) | 3 |
| float(x) | Float | float("5") | 5.0 |
| str(x) | String | str(25) | "25" |
| complex(x, y) | Complex | complex(2, 3) | (2+3j) |